```
!wget -O "dataset.zip" "https://data.mendeley.com/public-files/datasets/tywbtsjrjv/files/b4e3a32f-c0bd-4060-81e9
```

```
--2026-04-30 10:18:02--  https://data.mendeley.com/public-files/datasets/tywbtsjrjv/files/b4e3a32f-c0bd-4060-81e9
Resolving data.mendeley.com (data.mendeley.com)... 162.159.130.86, 162.159.133.86
Connecting to data.mendeley.com (data.mendeley.com)|162.159.130.86|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://prod-dcd-datasets-public-files-eu-west-1.s3.eu-west-1.amazonaws.com/349ac012-2948-4172-bbba-3bf
--2026-04-30 10:18:03--  https://prod-dcd-datasets-public-files-eu-west-1.s3.eu-west-1.amazonaws.com/349ac012-294
Resolving prod-dcd-datasets-public-files-eu-west-1.s3.eu-west-1.amazonaws.com (prod-dcd-datasets-public-files-eu-
Connecting to prod-dcd-datasets-public-files-eu-west-1.s3.eu-west-1.amazonaws.com (prod-dcd-datasets-public-files
HTTP request sent, awaiting response... 200 OK
Length: 949267727 (905M) [application/zip]
Saving to: ‘dataset.zip’

dataset.zip         100%[===================>] 905.29M  27.6MB/s    in 35s

2026-04-30 10:18:38 (26.0 MB/s) - ‘dataset.zip’ saved [949267727/949267727]
```

```
!unzip /content/dataset.zip
```

```
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (948)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (949)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (95).JI
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (950)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (951)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (952)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (953)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (954)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (955)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (956)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (957)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (958)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (959)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (96).JI
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (960)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (961)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (962)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (963)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (964)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (965)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (966)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (967)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (968)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (969)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (97).JI
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (970)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (971)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (972)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (973)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (974)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (975)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (976)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (977)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (978)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (979)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (98).JI
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (980)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (981)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (982)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (983)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (984)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (985)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (986)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (987)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (988)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (989)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (99).JI
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (990)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (991)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (992)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (993)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (994)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (995)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (996)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (997)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (998)..
  inflating: Plant_leave_diseases_dataset_with_augmentation/Tomato___Tomato_Yellow_Leaf_Curl_Virus/image (999)..
```

```
import matplotlib.pyplot as plt
import numpy as np
import os
import tensorflow as tf
```

```
!pip install split-folders
```

```
Collecting split-folders
  Downloading split_folders-0.6.1-py3-none-any.whl.metadata (11 kB)
Downloading split_folders-0.6.1-py3-none-any.whl (13 kB)
Installing collected packages: split-folders
Successfully installed split-folders-0.6.1
```

```
import splitfolders
splitfolders.ratio('/content/Plant_leave_diseases_dataset_with_augmentation', output="dataset", seed=1337, ratio
```

```
Copying files: 61486 files [00:16, 3726.01 files/s]
```

Data Preprocessing

```
train_dir = "/content/dataset/train"
validation_dir = "/content/dataset/val"

BATCH_SIZE = 32
IMG_SIZE = (160, 160)

train_dataset = tf.keras.utils.image_dataset_from_directory(train_dir,
                                                            shuffle=True,
                                                            batch_size=BATCH_SIZE,
                                                            image_size=IMG_SIZE)
```

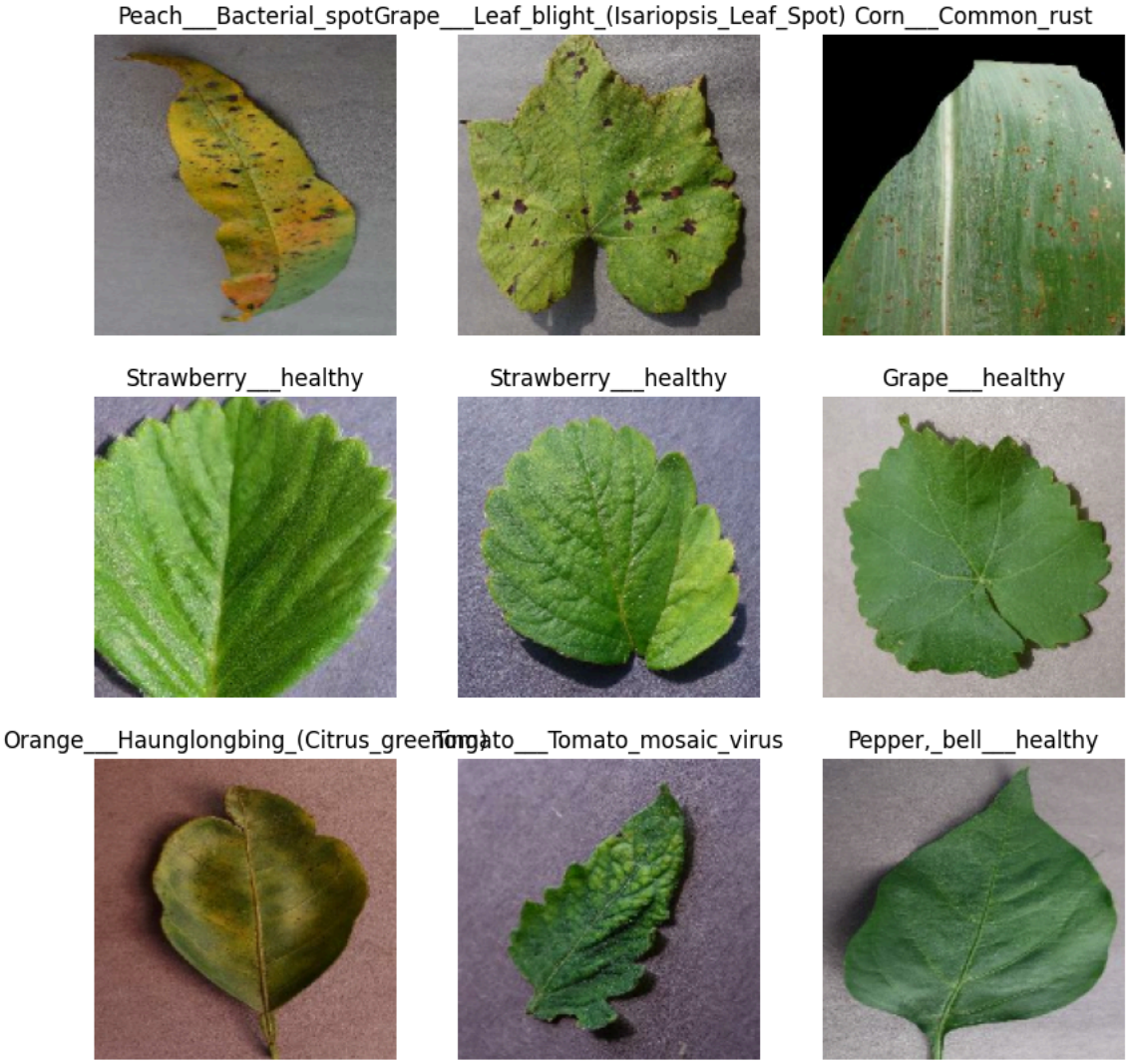
```
Found 49179 files belonging to 39 classes.
```

```
validation_dataset = tf.keras.utils.image_dataset_from_directory(validation_dir,
                                                                 shuffle=True,
                                                                 batch_size=BATCH_SIZE,
                                                                 image_size=IMG_SIZE)
test_dir = "/content/dataset/test"
test_dataset = tf.keras.utils.image_dataset_from_directory(test_dir,
                                                           batch_size=BATCH_SIZE,
                                                           image_size=IMG_SIZE)
```

```
Found 6139 files belonging to 39 classes.
Found 6168 files belonging to 39 classes.
```

```
class_names = train_dataset.class_names

plt.figure(figsize=(10, 10))
for images, labels in train_dataset.take(1):
  for i in range(9):
    ax = plt.subplot(3, 3, i + 1)
    plt.imshow(images[i].numpy().astype("uint8"))
    plt.title(class_names[labels[i]])
    plt.axis("off")
```




```
class_names
```

```
['Apple___Apple_scab',
 'Apple___Black_rot',
 'Apple___Cedar_apple_rust',
 'Apple___healthy',
 'Background_without_leaves',
 'Blueberry___healthy',
 'Cherry___Powdery_mildew',
 'Cherry___healthy',
 'Corn___Cercospora_leaf_spot Gray_leaf_spot',
 'Corn___Common_rust',
 'Corn___Northern_Leaf_Blight',
 'Corn___healthy',
 'Grape___Black_rot',
 'Grape___Esca_(Black_Measles)',
 'Grape___Leaf_blight_(Isariopsis_Leaf_Spot)',
 'Grape___healthy',
 'Orange___Haunglongbing_(Citrus_greening)',
 'Peach___Bacterial_spot',
 'Peach___healthy',
 'Pepper,_bell___Bacterial_spot',
 'Pepper,_bell___healthy',
 'Potato___Early_blight',
 'Potato___Late_blight',
 'Potato___healthy',
 'Raspberry___healthy',
 'Soybean___healthy',
 'Squash___Powdery_mildew',
 'Strawberry___Leaf_scorch',
 'Strawberry___healthy',
 'Tomato___Bacterial_spot',
 'Tomato___Early_blight',
 'Tomato___Late_blight',
 'Tomato___Leaf_Mold',
 'Tomato___Septoria_leaf_spot',
 'Tomato___Spider_mites Two-spotted_spider_mite',
 'Tomato___Target_Spot',
 'Tomato___Tomato_Yellow_Leaf_Curl_Virus',
 'Tomato___Tomato_mosaic_virus',
 'Tomato___healthy']
```

```
AUTOTUNE = tf.data.AUTOTUNE

train_dataset = train_dataset.prefetch(buffer_size=AUTOTUNE)
validation_dataset = validation_dataset.prefetch(buffer_size=AUTOTUNE)
test_dataset = test_dataset.prefetch(buffer_size=AUTOTUNE)
```

```
preprocess_input = tf.keras.applications.efficientnet.preprocess_input
```

Create the base model from the pre-trained convnets

```
IMG_SHAPE = IMG_SIZE + (3,)
base_model = tf.keras.applications.EfficientNetB4(
    input_shape=IMG_SHAPE,
    include_top=False,
    weights='imagenet',
)
```

```
Downloading data from https://storage.googleapis.com/keras-applications/efficientnetb4_notop.h5
71686520/71686520 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
image_batch, label_batch = next(iter(train_dataset))
feature_batch = base_model(image_batch)
print(feature_batch.shape)
```

```
(32, 5, 5, 1792)
```

Feature extraction

Freeze the convolutional base

```
base_model.trainable = False
```

```
base_model.summary()
```

Model: "efficientnetb4"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 160, 160, 3)	0	–
rescaling (Rescaling)	(None, 160, 160, 3)	0	input_layer[0][0]
normalization (Normalization)	(None, 160, 160, 3)	7	rescaling[0][0]
rescaling_1 (Rescaling)	(None, 160, 160, 3)	0	normalization[0]…
stem_conv_pad (ZeroPadding2D)	(None, 161, 161, 3)	0	rescaling_1[0][0]
stem_conv (Conv2D)	(None, 80, 80, 48)	1,296	stem_conv_pad[0]…
stem_bn (BatchNormalizatio…	(None, 80, 80, 48)	192	stem_conv[0][0]
stem_activation (Activation)	(None, 80, 80, 48)	0	stem_bn[0][0]
block1a_dwconv (DepthwiseConv2D)	(None, 80, 80, 48)	432	stem_activation[…
block1a_bn (BatchNormalizatio…	(None, 80, 80, 48)	192	block1a_dwconv[0…
block1a_activation (Activation)	(None, 80, 80, 48)	0	block1a_bn[0][0]
block1a_se_squeeze (GlobalAveragePool…	(None, 48)	0	block1a_activati…
block1a_se_reshape (Reshape)	(None, 1, 1, 48)	0	block1a_se_squee…
block1a_se_reduce (Conv2D)	(None, 1, 1, 12)	588	block1a_se_resha…
block1a_se_expand (Conv2D)	(None, 1, 1, 48)	624	block1a_se_reduc…
block1a_se_excite (Multiply)	(None, 80, 80, 48)	0	block1a_activati… block1a_se_expan…
block1a_project_co… (Conv2D)	(None, 80, 80, 24)	1,152	block1a_se_excit…
block1a_project_bn (BatchNormalizatio…	(None, 80, 80, 24)	96	block1a_project_…
block1b_dwconv (DepthwiseConv2D)	(None, 80, 80, 24)	216	block1a_project_…
block1b_bn (BatchNormalizatio…	(None, 80, 80, 24)	96	block1b_dwconv[0…
block1b_activation (Activation)	(None, 80, 80, 24)	0	block1b_bn[0][0]
block1b_se_squeeze (GlobalAveragePool…	(None, 24)	0	block1b_activati…
block1b_se_reshape (Reshape)	(None, 1, 1, 24)	0	block1b_se_squee…
block1b_se_reduce (Conv2D)	(None, 1, 1, 6)	150	block1b_se_resha…
block1b_se_expand (Conv2D)	(None, 1, 1, 24)	168	block1b_se_reduc…
block1b_se_excite (Multiply)	(None, 80, 80, 24)	0	block1b_activati… block1b_se_expan…
block1b_project_co… (Conv2D)	(None, 80, 80, 24)	576	block1b_se_excit…
block1b_project_bn (BatchNormalizatio…	(None, 80, 80, 24)	96	block1b_project_…
block1b_drop (Dropout)	(None, 80, 80, 24)	0	block1b_project_…
block1b_add (Add)	(None, 80, 80, 24)	0	block1b_drop[0][… block1a_project_…
block2a_expand_conv (Conv2D)	(None, 80, 80, 144)	3,456	block1b_add[0][0]
block2a_expand_bn	(None, 80, 80,	576	block2a_expand_c…

(BatchNormalizatio…	144)		
block2a_expand_act… (Activation)	(None, 80, 80, 144)	0	block2a_expand_b…
block2a_dwconv_pad (ZeroPadding2D)	(None, 81, 81, 144)	0	block2a_expand_a…
block2a_dwconv (DepthwiseConv2D)	(None, 40, 40, 144)	1,296	block2a_dwconv_p…
block2a_bn (BatchNormalizatio…	(None, 40, 40, 144)	576	block2a_dwconv[0…
block2a_activation (Activation)	(None, 40, 40, 144)	0	block2a_bn[0][0]
block2a_se_squeeze (GlobalAveragePool…	(None, 144)	0	block2a_activati…
block2a_se_reshape (Reshape)	(None, 1, 1, 144)	0	block2a_se_squee…
block2a_se_reduce (Conv2D)	(None, 1, 1, 6)	870	block2a_se_resha…
block2a_se_expand (Conv2D)	(None, 1, 1, 144)	1,008	block2a_se_reduc…
block2a_se_excite (Multiply)	(None, 40, 40, 144)	0	block2a_activati… block2a_se_expan…
block2a_project_co… (Conv2D)	(None, 40, 40, 32)	4,608	block2a_se_excit…
block2a_project_bn (BatchNormalizatio…	(None, 40, 40, 32)	128	block2a_project_…
block2b_expand_conv (Conv2D)	(None, 40, 40, 192)	6,144	block2a_project_…
block2b_expand_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2b_expand_c…
block2b_expand_act… (Activation)	(None, 40, 40, 192)	0	block2b_expand_b…
block2b_dwconv (DepthwiseConv2D)	(None, 40, 40, 192)	1,728	block2b_expand_a…
block2b_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2b_dwconv[0…
block2b_activation (Activation)	(None, 40, 40, 192)	0	block2b_bn[0][0]
block2b_se_squeeze (GlobalAveragePool…	(None, 192)	0	block2b_activati…
block2b_se_reshape (Reshape)	(None, 1, 1, 192)	0	block2b_se_squee…
block2b_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block2b_se_resha…
block2b_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block2b_se_reduc…
block2b_se_excite (Multiply)	(None, 40, 40, 192)	0	block2b_activati… block2b_se_expan…
block2b_project_co… (Conv2D)	(None, 40, 40, 32)	6,144	block2b_se_excit…
block2b_project_bn (BatchNormalizatio…	(None, 40, 40, 32)	128	block2b_project_…
block2b_drop (Dropout)	(None, 40, 40, 32)	0	block2b_project_…
block2b_add (Add)	(None, 40, 40, 32)	0	block2b_drop[0][… block2a_project_…
block2c_expand_conv (Conv2D)	(None, 40, 40, 192)	6,144	block2b_add[0][0]
block2c_expand_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2c_expand_c…
block2c_expand_act… (Activation)	(None, 40, 40, 192)	0	block2c_expand_b…
block2c_dwconv (DepthwiseConv2D)	(None, 40, 40, 192)	1,728	block2c_expand_a…
block2c_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2c_dwconv[0…

block2c_activation (Activation)	(None, 40, 40, 192)	0	block2c_bn[0][0]
block2c_se_squeeze (GlobalAveragePool…	(None, 192)	0	block2c_activati…
block2c_se_reshape (Reshape)	(None, 1, 1, 192)	0	block2c_se_squee…
block2c_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block2c_se_resha…
block2c_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block2c_se_reduc…
block2c_se_excite (Multiply)	(None, 40, 40, 192)	0	block2c_activati… block2c_se_expan…
block2c_project_co… (Conv2D)	(None, 40, 40, 32)	6,144	block2c_se_excit…
block2c_project_bn (BatchNormalizatio…	(None, 40, 40, 32)	128	block2c_project_…
block2c_drop (Dropout)	(None, 40, 40, 32)	0	block2c_project_…
block2c_add (Add)	(None, 40, 40, 32)	0	block2c_drop[0][… block2b_add[0][0]
block2d_expand_conv (Conv2D)	(None, 40, 40, 192)	6,144	block2c_add[0][0]
block2d_expand_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2d_expand_c…
block2d_expand_act… (Activation)	(None, 40, 40, 192)	0	block2d_expand_b…
block2d_dwconv (DepthwiseConv2D)	(None, 40, 40, 192)	1,728	block2d_expand_a…
block2d_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block2d_dwconv[0…
block2d_activation (Activation)	(None, 40, 40, 192)	0	block2d_bn[0][0]
block2d_se_squeeze (GlobalAveragePool…	(None, 192)	0	block2d_activati…
block2d_se_reshape (Reshape)	(None, 1, 1, 192)	0	block2d_se_squee…
block2d_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block2d_se_resha…
block2d_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block2d_se_reduc…
block2d_se_excite (Multiply)	(None, 40, 40, 192)	0	block2d_activati… block2d_se_expan…
block2d_project_co… (Conv2D)	(None, 40, 40, 32)	6,144	block2d_se_excit…
block2d_project_bn (BatchNormalizatio…	(None, 40, 40, 32)	128	block2d_project_…
block2d_drop (Dropout)	(None, 40, 40, 32)	0	block2d_project_…
block2d_add (Add)	(None, 40, 40, 32)	0	block2d_drop[0][… block2c_add[0][0]
block3a_expand_conv (Conv2D)	(None, 40, 40, 192)	6,144	block2d_add[0][0]
block3a_expand_bn (BatchNormalizatio…	(None, 40, 40, 192)	768	block3a_expand_c…
block3a_expand_act… (Activation)	(None, 40, 40, 192)	0	block3a_expand_b…
block3a_dwconv_pad (ZeroPadding2D)	(None, 43, 43, 192)	0	block3a_expand_a…
block3a_dwconv (DepthwiseConv2D)	(None, 20, 20, 192)	4,800	block3a_dwconv_p…
block3a_bn (BatchNormalizatio…	(None, 20, 20, 192)	768	block3a_dwconv[0…
block3a_activation (Activation)	(None, 20, 20, 192)	0	block3a_bn[0][0]
block3a_se_squeeze (GlobalAveragePool…	(None, 192)	0	block3a_activati…

block3a_se_reshape (Reshape)	(None, 1, 1, 192)	0	block3a_se_squee…
block3a_se_reduce (Conv2D)	(None, 1, 1, 8)	1,544	block3a_se_resha…
block3a_se_expand (Conv2D)	(None, 1, 1, 192)	1,728	block3a_se_reduc…
block3a_se_excite (Multiply)	(None, 20, 20, 192)	0	block3a_activati… block3a_se_expan…
block3a_project_co… (Conv2D)	(None, 20, 20, 56)	10,752	block3a_se_excit…
block3a_project_bn (BatchNormalizatio…	(None, 20, 20, 56)	224	block3a_project_…
block3b_expand_conv (Conv2D)	(None, 20, 20, 336)	18,816	block3a_project_…
block3b_expand_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3b_expand_c…
block3b_expand_act… (Activation)	(None, 20, 20, 336)	0	block3b_expand_b…
block3b_dwconv (DepthwiseConv2D)	(None, 20, 20, 336)	8,400	block3b_expand_a…
block3b_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3b_dwconv[0…
block3b_activation (Activation)	(None, 20, 20, 336)	0	block3b_bn[0][0]
block3b_se_squeeze (GlobalAveragePool…	(None, 336)	0	block3b_activati…
block3b_se_reshape (Reshape)	(None, 1, 1, 336)	0	block3b_se_squee…
block3b_se_reduce (Conv2D)	(None, 1, 1, 14)	4,718	block3b_se_resha…
block3b_se_expand (Conv2D)	(None, 1, 1, 336)	5,040	block3b_se_reduc…
block3b_se_excite (Multiply)	(None, 20, 20, 336)	0	block3b_activati… block3b_se_expan…
block3b_project_co… (Conv2D)	(None, 20, 20, 56)	18,816	block3b_se_excit…
block3b_project_bn (BatchNormalizatio…	(None, 20, 20, 56)	224	block3b_project_…
block3b_drop (Dropout)	(None, 20, 20, 56)	0	block3b_project_…
block3b_add (Add)	(None, 20, 20, 56)	0	block3b_drop[0][… block3a_project_…
block3c_expand_conv (Conv2D)	(None, 20, 20, 336)	18,816	block3b_add[0][0]
block3c_expand_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3c_expand_c…
block3c_expand_act… (Activation)	(None, 20, 20, 336)	0	block3c_expand_b…
block3c_dwconv (DepthwiseConv2D)	(None, 20, 20, 336)	8,400	block3c_expand_a…
block3c_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3c_dwconv[0…
block3c_activation (Activation)	(None, 20, 20, 336)	0	block3c_bn[0][0]
block3c_se_squeeze (GlobalAveragePool…	(None, 336)	0	block3c_activati…
block3c_se_reshape (Reshape)	(None, 1, 1, 336)	0	block3c_se_squee…
block3c_se_reduce (Conv2D)	(None, 1, 1, 14)	4,718	block3c_se_resha…
block3c_se_expand (Conv2D)	(None, 1, 1, 336)	5,040	block3c_se_reduc…
block3c_se_excite (Multiply)	(None, 20, 20, 336)	0	block3c_activati… block3c_se_expan…
block3c_project_co… (Conv2D)	(None, 20, 20, 56)	18,816	block3c_se_excit…

(Conv2D)	56)		
block3c_project_bn (BatchNormalizatio…	(None, 20, 20, 56)	224	block3c_project_…
block3c_drop (Dropout)	(None, 20, 20, 56)	0	block3c_project_…
block3c_add (Add)	(None, 20, 20, 56)	0	block3c_drop[0][… block3b_add[0][0]
block3d_expand_conv (Conv2D)	(None, 20, 20, 336)	18,816	block3c_add[0][0]
block3d_expand_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3d_expand_c…
block3d_expand_act… (Activation)	(None, 20, 20, 336)	0	block3d_expand_b…
block3d_dwconv (DepthwiseConv2D)	(None, 20, 20, 336)	8,400	block3d_expand_a…
block3d_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block3d_dwconv[0…
block3d_activation (Activation)	(None, 20, 20, 336)	0	block3d_bn[0][0]
block3d_se_squeeze (GlobalAveragePool…	(None, 336)	0	block3d_activati…
block3d_se_reshape (Reshape)	(None, 1, 1, 336)	0	block3d_se_squee…
block3d_se_reduce (Conv2D)	(None, 1, 1, 14)	4,718	block3d_se_resha…
block3d_se_expand (Conv2D)	(None, 1, 1, 336)	5,040	block3d_se_reduc…
block3d_se_excite (Multiply)	(None, 20, 20, 336)	0	block3d_activati… block3d_se_expan…
block3d_project_co… (Conv2D)	(None, 20, 20, 56)	18,816	block3d_se_excit…
block3d_project_bn (BatchNormalizatio…	(None, 20, 20, 56)	224	block3d_project_…
block3d_drop (Dropout)	(None, 20, 20, 56)	0	block3d_project_…
block3d_add (Add)	(None, 20, 20, 56)	0	block3d_drop[0][… block3c_add[0][0]
block4a_expand_conv (Conv2D)	(None, 20, 20, 336)	18,816	block3d_add[0][0]
block4a_expand_bn (BatchNormalizatio…	(None, 20, 20, 336)	1,344	block4a_expand_c…
block4a_expand_act… (Activation)	(None, 20, 20, 336)	0	block4a_expand_b…
block4a_dwconv_pad (ZeroPadding2D)	(None, 21, 21, 336)	0	block4a_expand_a…
block4a_dwconv (DepthwiseConv2D)	(None, 10, 10, 336)	3,024	block4a_dwconv_p…
block4a_bn (BatchNormalizatio…	(None, 10, 10, 336)	1,344	block4a_dwconv[0…
block4a_activation (Activation)	(None, 10, 10, 336)	0	block4a_bn[0][0]
block4a_se_squeeze (GlobalAveragePool…	(None, 336)	0	block4a_activati…
block4a_se_reshape (Reshape)	(None, 1, 1, 336)	0	block4a_se_squee…
block4a_se_reduce (Conv2D)	(None, 1, 1, 14)	4,718	block4a_se_resha…
block4a_se_expand (Conv2D)	(None, 1, 1, 336)	5,040	block4a_se_reduc…
block4a_se_excite (Multiply)	(None, 10, 10, 336)	0	block4a_activati… block4a_se_expan…
block4a_project_co… (Conv2D)	(None, 10, 10, 112)	37,632	block4a_se_excit…
block4a_project_bn (BatchNormalizatio…	(None, 10, 10, 112)	448	block4a_project_…
block4b_expand_conv	(None, 10, 10,	75,264	block4a_project_…

(Conv2D)	672)		
block4b_expand_bn (BatchNormalizatio…	(None, 10, 10, 672)	2,688	block4b_expand_c…
block4b_expand_act… (Activation)	(None, 10, 10, 672)	0	block4b_expand_b…
block4b_dwconv (DepthwiseConv2D)	(None, 10, 10, 672)	6,048	block4b_expand_a…
block4b_bn (BatchNormalizatio…	(None, 10, 10, 672)	2,688	block4b_dwconv[0…
block4b_activation (Activation)	(None, 10, 10, 672)	0	block4b_bn[0][0]
block4b_se_squeeze (GlobalAveragePool…	(None, 672)	0	block4b_activati…
block4b_se_reshape (Reshape)	(None, 1, 1, 672)	0	block4b_se_squee…
block4b_se_reduce (Conv2D)	(None, 1, 1, 28)	18,844	block4b_se_resha…
block4b_se_expand (Conv2D)	(None, 1, 1, 672)	19,488	block4b_se_reduc…
block4b_se_excite (Multiply)	(None, 10, 10, 672)	0	block4b_activati… block4b_se_expan…
block4b_project_co… (Conv2D)	(None, 10, 10, 112)	75,264	block4b_se_excit…
block4b_project_bn (BatchNormalizatio…	(None, 10, 10, 112)	448	block4b_project_…
block4b_drop (Dropout)	(None, 10, 10, 112)	0	block4b_project_…
block4b_add (Add)	(None, 10, 10, 112)	0	block4b_drop[0][… block4a_project_…
block4c_expand_conv (Conv2D)	(None, 10, 10, 672)	75,264	block4b_add[0][0]
block4c_expand_bn (BatchNormalizatio…	(None, 10, 10, 672)	2,688	block4c_expand_c…
block4c_expand_act… (Activation)	(None, 10, 10, 672)	0	block4c_expand_b…
block4c_dwconv (DepthwiseConv2D)	(None, 10, 10, 672)	6,048	block4c_expand_a…
block4c_bn (BatchNormalizatio…	(None, 10, 10, 672)	2,688	block4c_dwconv[0…
block4c_activation (Activation)	(None, 10, 10, 672)	0	block4c_bn[0][0]
block4c_se_squeeze (GlobalAveragePool…	(None, 672)	0	block4c_activati…
block4c_se_reshape (Reshape)	(None, 1, 1, 672)	0	block4c_se_squee…
block4c_se_reduce (Conv2D)	(None, 1, 1, 28)	18,844	block4c_se_resha…
block4c_se_expand (Conv2D)	(None, 1, 1, 672)	19,488	block4c_se_reduc…
block4c_se_excite (Multiply)	(None, 10, 10, 672)	0	block4c_activati… block4c_se_expan…
block4c_project_co… (Conv2D)	(None, 10, 10, 112)	75,264	block4c_se_excit…
block4c_project_bn (BatchNormalizatio…	(None, 10, 10, 112)	448	block4c_project_…
block4c_drop (Dropout)	(None, 10, 10, 112)	0	block4c_project_…
block4c_add (Add)	(None, 10, 10, 112)	0	block4c_drop[0][… block4b_add[0][0]
block4d_expand_conv (Conv2D)	(None, 10, 10, 672)	75,264	block4c_add[0][0]
block4d_expand_bn (BatchNormalizatio…	(None, 10, 10, 672)	2,688	block4d_expand_c…
block4d_expand_act… (Activation)	(None, 10, 10, 672)	0	block4d_expand_b…